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Atmospheric noble gases as tracers of biogenic gas dynamics in a shallow unconfined aquifer

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Abstract

Atmospheric noble gases (NGs) were used to investigate biogenic gas dynamics in a shallow unconfined aquifer impacted by a crude oil spill, near Bemidji, MN. Concentrations of $^{3,4}He$, $^{20,22}Ne$, $^{36,40}Ar$, Kr, and Xe were determined for gas- and aqueous-phase samples collected from the vadose and saturated zones, respectively. Systematic elemental fractionation of Ne, Ar, Kr, and Xe with respect to air was observed in both of these hydrogeologic zones. Within the vadose zone, relative ratios of Ne and Ar to Kr and Xe revealed distinct process-related trends when compared to corresponding ratios for air. The degree of NG deviation from atmospheric concentrations generally increased with greater atomic mass (i.e., $\Delta Xe > \Delta Kr > \Delta Ar > \Delta Ne$), indicating that Kr and Xe are the most sensitive NG tracers in the vadose zone. Reactive transport modeling of the gas data confirms that elemental fractionation can be explained by mass-dependent variations in diffusive fluxes of NGs opposite to a total pressure gradient established between different biogeochemical process zones. Depletion of atmospheric NGs was also observed within a methanogenic zone of petroleum hydrocarbon degradation located below the water table. Solubility normalized NG abundances followed the order Xe > Kr > Ar > Ne, which is indicative of dissolved NG partitioning into the gas phase in response to bubble formation and possibly ebullition. Observed elemental NG ratios of Ne/Kr, Ne/Xe, Ar/Xe, and Kr/Xe and a modeling analysis provide strong evidence that CH_4 generation below the water table caused gas exsolution and possibly ebullition and carbon transfer from groundwater to the vadose zone. These results suggest that noble gases provide sensitive tracers in biologically active unconfined aquifers and can assist in identifying carbon cycling and transfer within the vadose zone, the capillary fringe, and below the water table.

1. INTRODUCTION

Biogenic gases are principal products of key terminal electron accepting processes in the environment. Carbon dioxide (CO_2) is generated via microbial respiration, where organic matter oxidation is coupled with the reduction of an electron acceptor. Predominant electron acceptors in soils, sediments, and aquifers include oxygen (O_2), nitrate (NO_3), manganese (Mn), iron (Fe), and sulfate (SO_4). Methane (CH_4) is produced in anoxic systems where methanogenesis becomes an important pathway for organic matter decomposition (Whiticar et al., 1986). Although rates of biogenic gas production and organic matter decomposition are coupled, measurements of CH_4 and CO_2 production rates are complicated by their subsurface reactivity (Sihota et al., 2011; Sihota and Mayer, 2012). For example, CH_4 can be utilized as a carbon source or electron donor (Hallam et al., 2004; Modin et al., 2007; Amos et al.,

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2012; Segarra et al., 2013), whereas CO_2 may be consumed by carbonate–mineral precipitation (Baedecker et al., 1993; Holden et al., 2013) or reduction to CH_4 in the presence of hydrogen (Whiticar et al., 1986). Concentrations and distributions of biogenic gases are also influenced by advective and diffusive transport in groundwater and soil gas (Amos et al., 2005). Although direct measurement of biogenic gas fluxes is possible, these data may provide incomplete information on their origin, production, fate, and transport. Development of improved techniques for constraining biogenic gas dynamics is critical for estimating rates of organic matter decomposition and, therefore, assessing carbon mass balances, including mass transfer processes across the water table, and the ground surface.

Atmospheric noble gases (NGs) are important tracers of physical and biological processes in lakes, rivers, groundwater, and soils (Kipfer et al., 2002; Mächler et al., 2012; Brennwald et al., 2013; Freundt et al., 2013). These inert gases are subject to the same advective and diffusive transport as CH_4 and CO_2 within these hydrogeologic systems, yet processes that control biogenic gas production do not directly influence their abundance. Meteoric water (i.e. water in equilibrium with the atmosphere) represents the principal and, for all practical purposes, only source of NGs in these systems. The initial NG signatures are representative of local recharge temperature, elevation, salinity, and capture effects from excess air formation (Aeschbach-Hertig et al., 1999; Kipfer et al., 2002; Aeschbach-Hertig and Solomon, 2013). Atmospheric NGs are therefore valuable tracers of recharge conditions in hydrogeologic studies (e.g., Manning and Solomon, 2003; Klump et al., 2008a,b; Althaus et al., 2009; Cey et al., 2009; Zhu and Kipfer, 2010; Grundl et al., 2013) and in paleoclimate investigations (e.g., Ballentine and Hall, 1999; Aeschbach-Hertig et al., 2000; Hall et al., 2012). Noble gases have also provided valuable insight into gas exchange in variably saturated porous media (Holocher et al., 2002; Klump et al., 2007, 2008a,b) and on CH_4 dynamics in lakes and oceans (Brennwald et al., 2005; Strassmann et al., 2005; Holzner et al., 2008). Although NGs are inert, their abundance and distribution are indirectly influenced by biogenic gas production and transport (Brennwald et al., 2005; Freundt et al., 2013). Therefore, the post-recharge evolution of NGs can potentially offer valuable insight into biogenic gas dynamics within the shallow subsurface, both above and below the water table.

Microbiological and hydrogeological processes are important controls on the abundance and distribution of gases within the vadose zone. Amos et al. (2005) used Ar and N_2, which also may exhibit limited reactivity under certain conditions, to investigate degradation of petroleum hydrocarbons in a shallow unconfined aquifer. Methanogenic and methanotrophic zones were delineated based on relative enrichment or depletion of these gases, respectively. Freundt et al. (2013) examined partial pressure responses of atmospheric NGs to total pressure gradients established by coupled microbial respiration and O_2 depletion in an unsaturated soil. Relative enrichment of NGs was observed in O_2 depletion zones and this effect was more pronounced for heavier NGs. This trend in NG enrichment was attributed to element-dependent differences of the physical properties of NG species during gas/water partitioning. Using atmospheric NGs, Mächler et al. (2013) identified the local partial dissolution of entrapped air in the quasi-saturated zone in response to water table fluctuations (i.e., induced excess air formation) as important to oxygenate groundwater and promote microbial respiration.

In addition, ebullition can represent a significant source of biogenic gas fluxes from lakes and wetlands (Delsontro et al., 2010; Schubert et al., 2012) and has been proposed as a key contributor to CH_4 effluxes from methanogenic groundwater systems (Amos et al., 2005; Amos and Mayer, 2006). Amos et al. (2005) interpreted observed Ar and N_2 depletions in the saturated zone of a shallow aquifer in combination with batch modeling and the observation of gas bubbles in cores, as evidence that gas exsolution and possibly ebullition comprised an important component of the CH_4 flux in this system. However, Ar and N_2 data alone were insufficient to confirm with certainty that ebullition actually occurred, or whether gas bubbles formed by exsolution remained entrapped due to capillary forces. Brennwald et al. (2005) measured NG abundances and isotopic ratios to examine CH_4 fluxes in methanogenic lake sediments. Pore waters exhibited extensive NG depletion immediately below the sediment–water interface. This depletion was most pronounced for Ne due to the positive relationship between atomic mass and solubility in water. Ratios of $^{20}Ne/^{22}Ne$ and $^{36}Ar/^{40}Ar$ agreed with the respective atmospheric ratios and were hence deemed indicative of equilibrium partitioning of NGs into the opening gas bubbles during formation and subsequent ebullition from the sediment matrix. Subsequent modeling of noble gas abundances constrained CH_4 fluxes and revealed that ebullition is the dominant mechanism of CH_4 release in this lacustrine system (Brennwald et al., 2005).

The current study utilized atmospheric NGs to examine biogenic gas dynamics in a shallow unconfined aquifer impacted by a crude oil spill. Abundances of $^{3,4}He$, $^{20,22}Ne$, $^{36,40}Ar$, Kr, and Xe were determined and used to examine vadose zone advective and diffusive transport of biogenic gases associated with methanogenic petroleum hydrocarbon degradation. Numerical modeling was performed to examine the physical controls on the co-evolution of CH_4 and noble gas abundances within this system. Noble gas data from the saturated zone was used to investigate the occurrence of gas exsolution, and possibly ebullition and gas transfer from the saturated into the vadose zone. To our knowledge, this study represents the first integrated application of NGs to examine biogenic gas dynamics in both the vadose zone and saturated zone of a shallow groundwater system.

2. SITE DESCRIPTION

Aqueous and gas-phase samples were collected from a shallow unconfined aquifer located approximately 16 km northwest of Bemidji, MN, USA. The site was contaminated in 1979 by approximately 1.7×10^6 L of crude oil following the rupture of a high-pressure underground pipeline. This glacial outwash aquifer is comprised of sandy gravel,

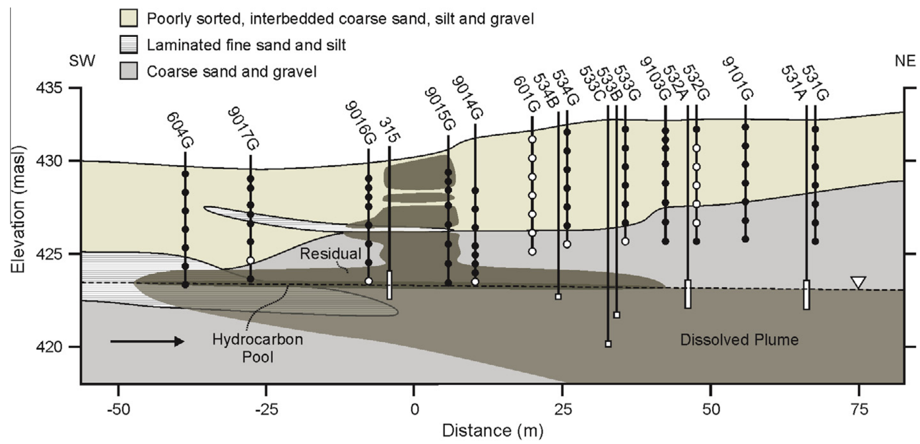

Fig. 1. Cross-sectional diagram showing distribution of vadose-zone wells and saturated-zone wells, local hydrogeologic conditions (after Smith and Hult, 1993), and the distribution of hydrocarbons in the vadose zone (after Dillard et al., 1997) and saturated zone (after Amos et al., 2005). Closed circles represent vadose zone gas wells sampled for major component gases, while open circles represent locations that were also sampled for noble gases. The extent of the well screens used for the collection of saturated zone samples are depicted as open rectangles. The direction of groundwater flow is indicated by the arrow, and the water table is defined by the dashed line.

gravely sand, and moderately- to poorly-sorted sand. Discontinuous sand and silt lenses occur throughout the site, and clayey till situated approximately 20 m below ground surface acts as a lower confining unit (Smith and Hult, 1993; Dillard et al., 1997). Annual recharge ranges from 0.1 to 0.3 m a^{-1} (Essaid et al., 2011) and the water table is situated 6 to 9 m below ground surface with annual fluctuations averaging 0.5 m (Amos et al., 2005). Detailed descriptions of hydrogeological and geochemical aspects of the study site can be found in Bennett et al. (1993), Baedcecker et al. (1993), Bekins et al. (1999, 2001), Cozzarelli et al. (2001) and Essaid et al. (2003, 2011).

Approximately 25% of the crude oil migrated into the subsurface, and two distinct subsurface oil bodies – the south pool and the north pool – subsequently developed. This study utilized an extensive network of monitoring wells located along a SW to NE transect along the axis of the north pool (Fig. 1). The north pool originally contained between 8 and 16 $\times$ 10^4 L of oil. The majority of this oil migrated to the water table, while the remainder was retained in the unsaturated zone. Dillard et al. (1997) reported that residual hydrocarbon saturation within the north pool ranged from 10% to 20 % in the vadose zone, and from 30% to 70 % in the uppermost 1 m of the saturated zone. Dissolved CH_4 concentrations as high as 15 mg L^{-1}, which represents >60% of saturation in water, have been reported in within the dissolved hydrocarbon plume that emanates from the north pool (Cozzarelli et al., 1994).

3. METHODOLOGY

3.1. Sample collection and analyses

3.1.1. Vadose zone

Gas samples were collected from multi-level vadose-zone wells located both within (9014G, 9015G, 9016G) and adjacent to (531G, 532G, 533G, 534G, 601G, 604G, 9017G) the original source zone, and from a background well (310G) located >200 m from the source zone (Fig. 1). The United States Geological Survey (USGS) constructed these wells as bundles of five to nine stainless steel tubes with outer diameters measuring either 0.032 m or 0.064 m and 0.1 m long screens positioned at their base. Sampling points were distributed at approximately 0.5 m intervals between the ground surface and the top of the capillary fringe.

Samples for quantification of major component gases (i.e., CH_4, CO_2, O_2 and Ar) were collected at 92 discrete locations (Fig. 1). The gas-phase samples were collected into 10 mL Pressure Lok glass syringes (VICI Valco Instruments Co. Inc.). The syringes were purged once and samples were collected maintaining a positive pressure to avoid contamination with atmospheric gases. These samples were analyzed immediately using a field portable dual channel gas chromatograph (GC; CP-4900, Varian, Inc.) equipped with thermal conductivity detectors. This GC uses He as the carrier gas, a MolSieve 5 Å column for Ar, O_2 and N_2 peak separation, and a PoraPLOT U column for CO_2 and CH_4 peak separation. Ambient air and gas calibration standards (Supelco Analytical, Sigma–Aldrich Co. LLC) were used to develop multipoint calibration curves, and sample gas mole fractions were determined by least-squares linear regression of corresponding peak areas.

Samples for noble gas analyses were collected at 19 discrete locations, which included two depth profiles (601G and 532G) located near the center and at the fringe of the contaminated region. These samples were collected into 9.5 mm diameter Cu tubing measuring 1 m long and fitted at one end with a check valve. The opposite end of the Cu tubing was connected to vapor wells with low-permeability Viton tubing using compression or barbed fittings (Swagelok Co., USA) and tubing clamps to prevent leaks. The Viton tubing was routed through the head of a peristaltic pump and gas was pumped through the Cu tubing at a rate of 25 mL s^{-1} until three tube volumes had passed through the check valve. With the pump running, refrigeration tubing clamps were used to seal the Cu tubing, first at the outflow end and subsequently at the inflow end resulting in a positive relative gas pressure within the tube. The

Cu tubing was then disconnected from the Viton tubing and crimp-sealed at both ends to prevent leaks during transport.

3.1.2. Saturated zone

Groundwater samples were collected from monitoring wells located within (315, 534B and 533B) and below (533C) the north pool, and also upgradient (310E) and downgradient (532A) of this zone (Fig. 1). Samples were collected into 1 m lengths of 9.5 mm diameter Cu tubing. This Cu tubing was connected to dedicated polyvinyl chloride (PVC) well tubing using Viton tubing with barbed fittings (Swagelok Co., USA) and tubing clamps. Groundwater was pumped through the Cu tubing at a rate of 0.5 L s^{-1} with down-hole positive displacement pumps (Geotechnical Services Inc., USA) while visually ensuring that inline degassing was not occurring during sampling. After three well volumes were purged through the Cu tubing, the Cu tubes were crimp-sealed by first closing the outflow and then the inlet end. This approach was used to achieve over pressurization of the water sample to minimize potential for degassing (Beyerle et al., 2000). Dissolved concentrations of major component gases have previously been reported for the study site (Amos et al., 2005), and were not determined during the present study.

3.1.3. Noble gas analyses

Noble gas analyses were carried out at the Noble Gas Laboratory at the Swiss Federal Institute of Technology (Zürich, Switzerland) using the well-established protocols for the high precision determination of noble gases in water and gas samples (Beyerle et al., 2000; Kipfer et al., 2002; Huber et al., 2006). After extracting gases from gas or water samples, and removing the reactive gas components by selective freezing and absorption by getters, noble gases were separated into He–Ne and Ar–Kr–Xe fractions and quantified using a two-step protocol.

In the first step, the He–Ne fraction was split into two aliquots. To analyze ^{4}He and ^{20,22}Ne the first aliquot was expanded into and analyzed in a non-commercial 90° magnetic sector mass spectrometer equipped with a Bauer-Signer source tuned for maximum linearity with regard to total gas pressure. To determine the ^{3}He/^{4}He ratio, the second aliquot was expanded to a commercial VG 5400 mass spectrometer whose ion source was tuned such that the determination of the isotopic composition of He is virtually independent of total gas pressure.

In a second step, the Ar–Kr–Xe fraction was diluted and expanded to the non-commercial mass spectrometer for final ^{36,40}Ar, Kr and Xe analysis. Overall precision for noble gas concentrations (He–Xe) is <1.5%, for ^{20}Ne/^{22}Ne and ^{36}Ar/^{40}Ar <0.5% and for ^{3}He/^{4}He <1% (see Tables 1 and 2). For further experimental details refer to Beyerle et al. (2000), Kipfer et al. (2002), and Huber et al. (2006).

3.2. Numerical modeling

Vertical gas transport at vadose zone well (601G) was simulated using the multicomponent reactive transport code MIN3P (Mayer et al., 2002) amended with the Dusty Gas Model (DGM) formulation for gas diffusion and advection (Molins and Mayer, 2007). The DGM describes diffusive transport using the Stefan-Maxwell equations (Mason and Malinauskas, 1983) to overcome the limitations of Fick's law, which describes single component diffusion and is only valid for gases at low partial pressures. Knudsen diffusion, multi-component diffusion and

Table 1
Noble gas concentrations for gas-phase samples collected from vadose-zone wells and an air sample collected at the study site. The average analytical error of noble gas concentrations was <1.5% (see main text).

Well	Depth[a] (m)	Concentration (cm^3 cm^{-3})				
ID		He × 10^{-6}	Ne × 10^{-5}	Ar × 10^{-3}	Kr × 10^{-6}	Xe × 10^{-8}
Air	–	5.34	1.83	9.41	1.14	8.64
310G	6.5	5.38	1.90	9.69	1.19	9.23
9017G	4.9	5.16	1.76	8.87	1.06	8.22
9016G	6.0	5.15	1.75	8.60	1.02	6.97
9014G	7.9	4.82	1.59	7.39	0.832	5.77
601G	0.5	6.02	1.83	9.40	1.14	8.81
	1.5	5.81	1.86	9.21	1.15	8.67
	2.5	5.78	–	9.41	1.12	8.46
	3.5	5.38	1.92	10.1	1.24	9.70
	4.5	5.30	1.84	9.34	1.15	8.66
	5.5	5.10	1.68	8.51	1.02	7.36
	6.5	4.93	1.62	7.53	0.853	6.04
534G	6.5	5.09	1.68	8.21	0.964	7.02
533G	5.5	2.57	0.922	4.81	0.594	4.51
	6.5	5.45	1.89	9.68	1.23	9.42
532G	1.5	6.33	1.80	9.70	1.19	9.15
	2.5	4.85	1.78	9.76	1.21	9.18
	3.5	4.91	1.86	10.2	1.27	9.31
	4.5	5.41	1.90	9.99	1.23	9.49
	5.5	5.24	1.79	9.49	1.16	8.62

[a] Median depth of screened interval below ground surface.

Table 2
Noble gas concentrations for groundwater samples collected from saturated-zone wells. The average analytical error of dissolved noble gas concentrations was <1.5% (see main text).

Well ID	Depth[a] (m)	Concentration ($cm^3_{STP}g^{-1}$) He $\times 10^{-8}$	Ne $\times 10^{-7}$	Ar $\times 10^{-4}$	Kr $\times 10^{-8}$	Xe $\times 10^{-8}$
310E	9.7	4.27	1.89	4.03	9.57	1.45
315	5.1	1.18	0.398	0.824	4.26	0.675
534B	9.2	2.12	0.692	1.73	4.43	0.756
533B	10.7	9.91	1.49	3.21	7.85	1.28
533C	12.2	3.61	1.61	3.87	9.40	1.49
532A	8.5	1.17	0.444	1.68	4.69	0.840
531A	8.8	8.31	2.85	3.12	6.59	0.920

[a] Median depth of screened interval below ground surface.

advection are represented by a series of non-linear equations described by Thorstenson and Pollock (1989), Massmann and Farrier (1992), and Molins and Mayer (2007). The formulation of the model and its verification are described in detail in Molins and Mayer (2007). The model was previously used to simulate vadose zone gas dynamics at the Bemidji site (Molins and Mayer, 2007; Molins et al., 2010, and Sihota and Mayer, 2012) and was here extended to include the full suite of noble gases.

The one-dimensional (vertical) domain was spatially discretized using the finite volume method; 40 cells were used to provide a resolution that adequately allowed reproducing observed concentration gradients of reactive and inert gases. The physical properties of the model domain were based upon previous field observations and modeling studies (Smith and Hult, 1993; Amos et al., 2005; Molins et al., 2010). The current model included a 1 m thick fine-grained unit located 4 m below the ground surface. Porosity was 0.40 in the upper 4 m and 0.33 in the fine-grained unit and the underlying sediments. Simulated pore water saturations ranged between 0.2 near the ground surface and 0.5 near the base of the domain and within the fine-grained unit, in-line with observations by Dillard et al. (1997). In-situ total CO_2 and CH_4 gas generation rates attributed to biodegradation reactions were constrained by measured CO_2 effluxes at the ground surface, corrected for background soil respiration (Sihota et al., 2011). Binary diffusion coefficients were calculated for all reactive gases (O_2, N_2, CO_2, CH_4) and noble gases (He, Ne, Ar, Kr, Xe) based on the Chapman Enskog theory (see Molins et al., 2010). The required Lennard-Jones potentials and viscosities for noble gases were calculated based upon data reported by Reid et al. (1977). Additional model input parameters, including binary diffusion coefficients for all gases, are supplied as Supplementary data.

The evolution of dissolved noble gas concentrations during methanogenesis, bubble formation and ebullition in the saturated zone were simulated using the geochemical code PHREEQCi (v3.06.7757) with the Lawrence Livermore National Laboratory (LLNL) database (Parkhurst and Appelo, 2013). The GAS-PHASE and REACTION options in PHREEQCi were utilized to simulate the evolution of noble gas composition for constant pressure under conditions of (a) gas exsolution and bubble formation and (b) gas exsolution, bubble formation and periodic gas bubble release via ebullition. Based on the elevation of the field site, a total pressure of 0.95 atm was assigned to the gas phase. Modeled noble gas concentrations in the saturated zone (water samples) were normalized to concentrations of air saturated water (ASW) calculated for the given mean temperature (8 °C) and mean salinity (5.4‰) of the groundwater, and the mean atmospheric pressure of the area. A detailed account of the PHRREQCi model set-up and the input parameters is supplied as Supplementary data.

For both models, Henry's constants for NGs were based upon values reported by Benson and Krause (1976) and Wilhelm et al. (1977).

4. RESULTS

4.1. Vadose zone

Elevated CH_4 concentrations were observed along an 80 m section of the north pool transect (Fig. 2) spanning vapor wells 604G to 533G (Fig. 1). The highest CH_4 concentrations, of 15–20% (v/v), were observed 1 to 2 m above the water table and corresponded to a zone of residual hydrocarbons located immediately above the north pool. Distinct gradients in CH_4 concentrations were observed above this zone, with decreases from >15% to <1% (v/v) occurring over vertical distances of 2–4 m. Abrupt declines in O_2 concentrations from near atmospheric values to <1% (v/v) were observed near the outer margin of the elevated CH_4 zone. Concentrations of CO_2 declined from >10 to <2% (v/v) in this zone; however, the CO_2 gradient was opposite to the direction of the O_2 gradient. Near atmospheric O_2 concentrations extended >5 m below ground surface at locations more distal to the residual hydrocarbons. Areas of relative N_2 enrichment were observed in the zone of CH_4 and O_2 removal, whereas relative N_2 depletion was associated with high CH_4 concentrations. These trends in the distribution of CH_4, O_2 and N_2 indicate that CH_4 production occurs in the region above the hydrocarbon pool, whereas a zone of CH_4 removal developed at vertical distances of 2–4 m closer to the ground surface (Fig. 3). The observed spatial CH_4 distribution is in agreement with results of previous field studies (Amos et al., 2005) and modeling exercises (Molins et al., 2010).

Noble gas concentrations exhibited considerable variability among sample locations; however, consistent trends

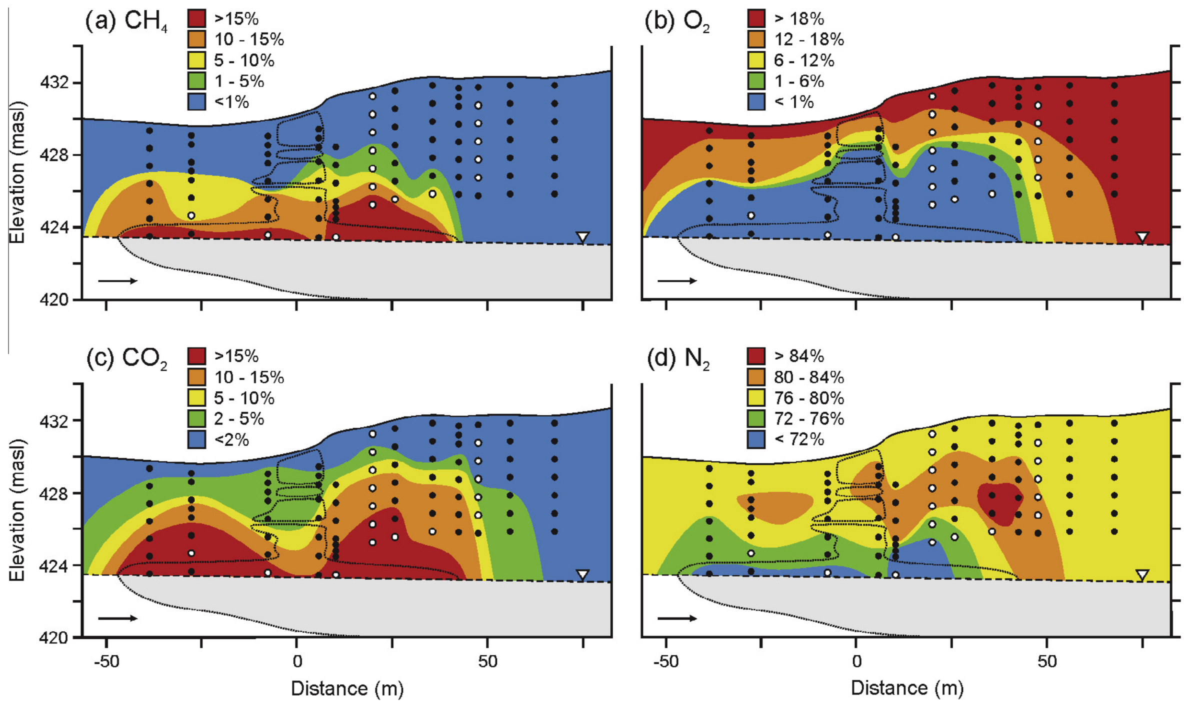

Fig. 2. Cross-sectional contour diagrams of volumetric concentrations (v/v) of major component gases in the vadose zone for July 2008. Closed circles represent vadose-zone wells sampled for major component gases, while open circles represent those also sampled for noble gases. Dotted contours represent distribution of residual hydrocarbons, while the gray shaded area denotes the dissolved hydrocarbon plume. The arrow indicates the direction of groundwater flow.

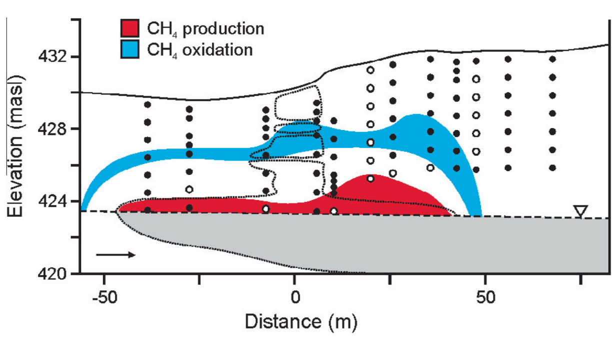

Fig. 3. Cross-sectional diagram showing inferred zones of methane production and removal. Closed circles represent vadose-zone wells sampled for major component gases, while open circles represent those also sampled for noble gases. Dotted contours represent distribution of residual hydrocarbons, while the gray shaded area denotes the dissolved hydrocarbon plume. The direction of groundwater flow is indicated by an arrow.

in Ne, Ar, Kr and Xe abundances at given locations were evident (Table 1). Concentrations of Ne, Ar, Kr and Xe within the methanogenic zone were generally depleted relative to atmospheric values (Fig. 4, vapor well 601G). Samples collected from vapor wells positioned nearest to the water table, where CH_4 concentrations commonly exceeded 15% (v/v), consistently exhibited the lowest noble gas concentrations. Air normalized depletion of noble gases increased with atomic mass in the zone of CH_4 production. Noble gas depletion in relation to atmospheric levels therefore followed the general order $C_{Xe}/C_{Xe^*} > C_{Kr}/C_{Kr^*} > C_{Ar}/C_{Ar^*} > C_{Ne}/C_{Ne^*}$ within this zone (asterisks denote the atmospheric concentration of a given noble gas). Approximately 2 m above the water table (601G), where CH_4 concentrations exceeded 20% (v/v), the lowest C_{Xe}/C_{Xe^*} value of 0.69 was observed. The corresponding air-normalized ratios of Ne, Ar and Kr were 0.89, 0.81 and 0.75, respectively (Fig. 4). Mass-dependent elemental fractionation was also observed in samples collected from vapor wells located both adjacent to the residual hydrocarbon source (9016G, 9014G) and approximately 25 m downgradient of the source (534G). Elemental fractionation was most apparent from ratios of $(C_{Ne}/C_{Xe})/(C_{Ne^*}/C_{Xe^*})$ and $(C_{Ne}/C_{Kr})/(C_{Ne^*}/C_{Kr^*})$, which exceeded 1.3 for vapor wells located within the methanogenic zone and immediately above the water table (9014G, 9016G). Corresponding ratios of Ne to Ar, Ar to Kr and Xe, and Kr to Xe were also found to be elevated within this zone, but to a lesser extent. Scatter plots of these ratios reveal distinct and systematic trends in elemental fractionation of noble gases associated with CH_4 production (Fig. 5).

In contrast to trends observed for the zone of CH_4 production, noble gas concentrations associated with CH_4 oxidation were enriched relative to atmospheric values (Table 1). Values of C_{Xe}/C_{Xe^*} of 1.09 and 1.12 were observed above the residual hydrocarbons (601G) and above the dissolved hydrocarbon pool (532G), respectively (Fig. 4). Corresponding C_{Ne}/C_{Ne^*}, C_{Ar}/C_{Ar^*}, and C_{Kr}/C_{Kr^*} values were also enriched, but to a lesser extent than for C_{Xe}/C_{Xe^*}. Elemental fractionation of NGs at these locations exhibited a smaller magnitude compared to fractionation observed in the methanogenic zone, but deviations from

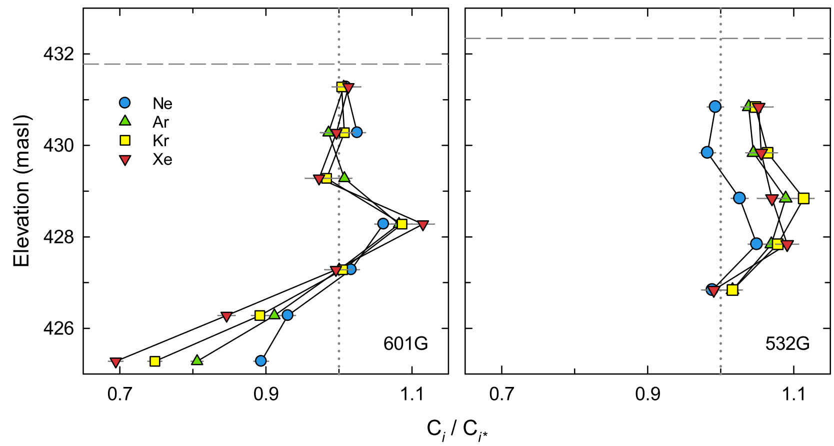

Fig. 4. Depth profiles of noble gas concentrations for gas-phase samples collected from vadose-zone wells 601G and 532G. Values are normalized to atmospheric concentrations, which are denoted by the dotted line.

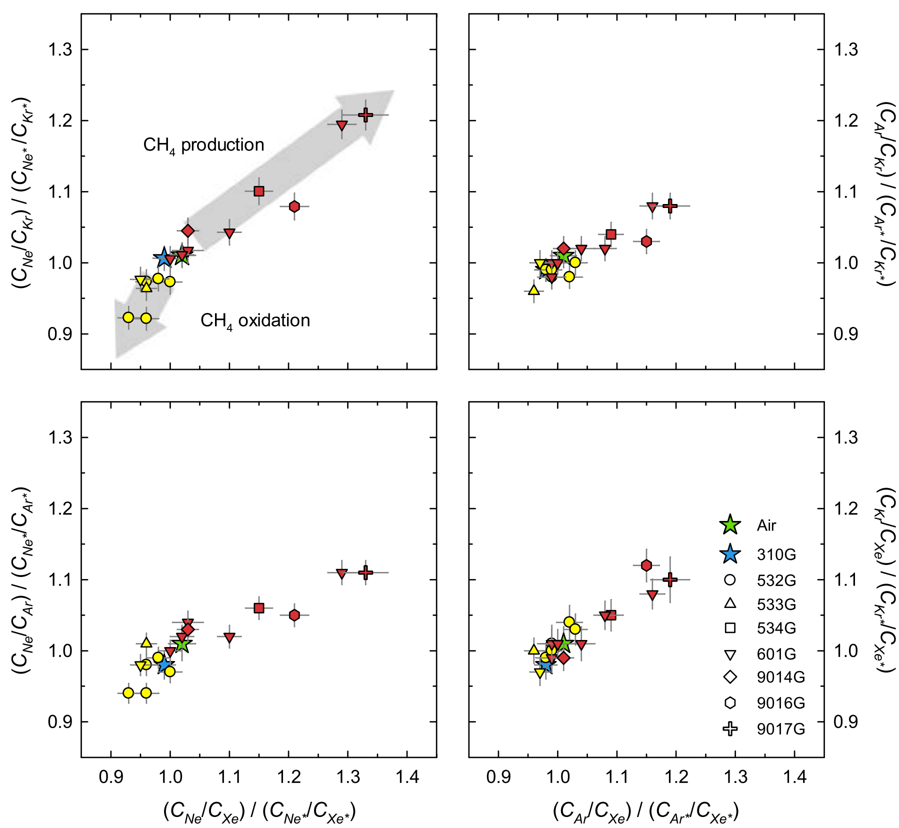

Fig. 5. Ratios of measured NG abundances normalized to corresponding atmospheric ratios (*). Observed trends in normalized NG ratios are categorized according to the predominant mode of carbon metabolism identified for a given vapor well. Symbol shape reflects the well location, red colored symbols are located within or in direct vicinity of the zone of CH_4 generation, and yellow colored symbols are located within the zone of CH_4 oxidation or above.

atmospheric levels were also most pronounced for the heavy noble gases (i.e., $C_{Xe}/C_{Xe^*} > C_{Kr}/C_{Kr^*} > C_{Ar}/C_{Ar^*} > C_{Ne}/C_{Ne^*}$). This trend was isolated to depths 4.5 m and 3.5 m at vapor wells 601G and 532G, respectively, which are located within the zone of CH_4 oxidation. Scatter plots of the respective normalized NG concentrations show a lesser elemental fractionation in other wells that exhibited NG enrichment (Fig. 5).

Trends in He concentrations relative to atmospheric NGs were rather inconsistent (Tables 1 and 2), suggesting input from non-atmospheric sources including accumulation of terrigenic He (Pinti and Bernard, 1995;

Solomon et al., 1996; Allard et al., 1997) or tritiogenic He (Schlosser et al., 1989; Aeschbach-Hertig, 1998; Kipfer et al., 2002; Manning et al., 2005). Due to the inherent non-systematic variability the He data are omitted from this analysis and subsequent discussion.

4.2. Saturated zone

Amos et al. (2005) conducted a comprehensive study of dissolved major component gases at the Bemidji site. High concentrations of CH_4 and CO_2, coupled with low O_2 concentrations, were observed within the dissolved hydrocarbon plume in 2002 and 2003. Methane concentrations of up to 5 mg L^{-1} were observed immediately below the hydrocarbon pool, but declined to <1 mg L^{-1} approximately 80 m downgradient. Dissolved CO_2 concentrations of up to 70 mg L^{-1} persisted further downgradient, whereas groundwater remained anoxic (O_2 < 0.1 mg L^{-1}) more than 50 m downgradient of the center of residual hydrocarbon mass.

Dissolved NG concentrations varied substantially among wells positioned along the axis of the dissolved hydrocarbon plume (Table 2). The upgradient background well 310E, exhibited NG abundances close to expected atmospheric ASW equilibrium concentrations, where relative saturation varied between 0.95 (Ne) and 1.05 (Xe). Dissolved NG concentrations were substantially lower than the respective ASW concentrations in samples from wells positioned both below and downgradient of the hydrocarbon pool, with the exception of well 531A. The normalized saturation ($C_i/C_{i,ASW}$) was generally larger for heavier noble gases (i.e., $C_{Xe}/C_{Xe,ASW} > C_{Kr}/C_{Xe,ASW} > C_{Ar}/C_{Ar,ASW} > C_{Ne}/C_{Ne,ASW}$), indicating that the lighter noble gases showed the most substantial deviations from ASW. Large NG depletion was apparent in wells positioned nearest to the water table and contaminant source zone (315, 532A, 534B). Normalized Ne saturations varied between 0.21 and 0.42 and those for Xe ranged from 0.49 to 0.61. Noble gas depletion was less pronounced in wells located further below the water table (533B, 533C). Dissolved NG concentrations exhibited an inverse trend in elemental fractionation at a distance of approximately 70 m downgradient (531A) of the residual hydrocarbon source, most likely due to excess air entrapment. Similar to the unsaturated zone, ratios of lighter (Ne, Ar) to heavier (Kr, Xe) NGs exhibited distinct trends (Fig. 6). Solubility-normalized ratios confirmed elemental fractionation with groundwater becoming more depleted in the lighter NGs.

5. DISCUSSION

5.1. Gas dynamics in the vadose zone

The observed NG abundances are indicative of spatial variations in total gas pressures associated with CH_4 production and oxidation. Substantial depletion of NGs relative to air was observed within the methanogenic zone. High CH_4 concentrations within this zone have previously

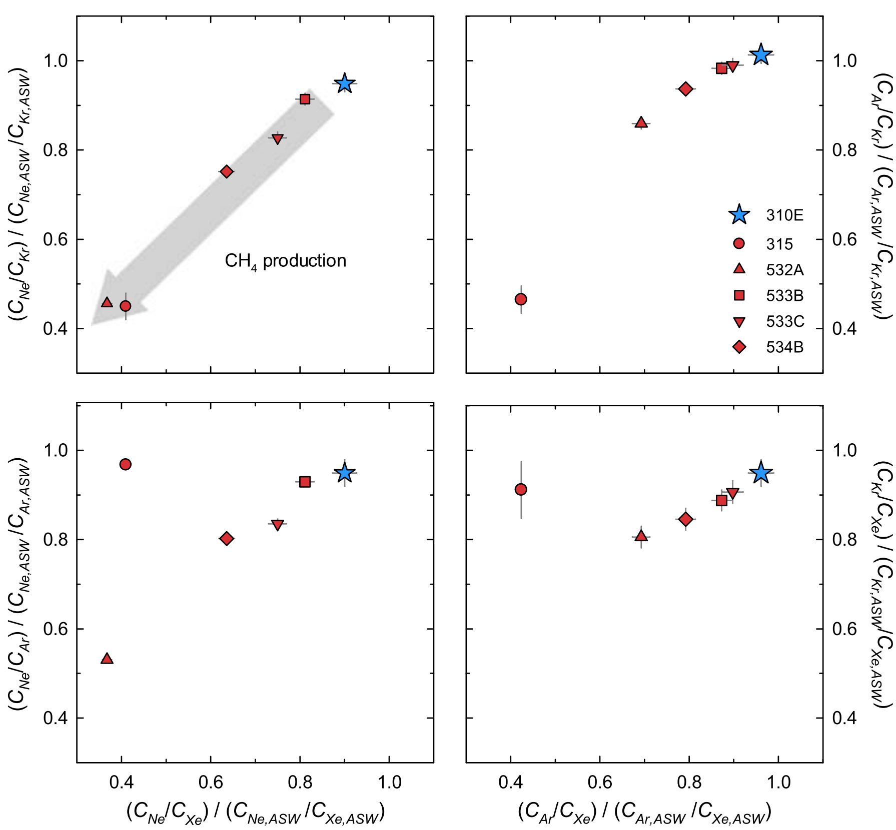

Fig. 6. Noble gas ratios calculated for groundwater samples collected from saturated-zone wells. Ratios are normalized to corresponding ratios of theoretical noble gas solubilities calculated for a given sample location. Ratios for 531A fall outside of the range used for these plots.

been attributed to methanogenesis via organic matter (CH_2O) fermentation (Revesz et al., 1995), where acetate and other organic acids serve as the terminal electron acceptor:

$$2CH_2O_{(aq)} \rightarrow CH_{4(g)} + CO_{2(g)} \quad (1)$$

This pathway generates both CH_4 and CO_2 and, although CO_2 exhibits relatively high solubility in water, field measurements (Amos et al., 2005) and reactive transport modeling (Molins et al., 2010) suggest that gas generation leads to increased total gas pressures within the methanogenic zone. In contrast, removal of CH_4 and O_2 via aerobic oxidation generates CO_2 and may lead to a decrease in total gas pressure:

$$CH_{4(g)} + 2O_{2(g)} \rightarrow CO_{2(g)} + 2H_2O_{(l)} \quad (2)$$

The relatively high solubility of CO_2 in water may further enhance the pressure decrease associated with this process. The net effect of these spatially distinct zones of CH_4 production and removal is a reaction-induced total gas pressure gradient (Amos et al., 2005; Molins et al., 2010).

All gases present in the vadose zone are subject to advective transport along this reaction-induced pressure gradient. Variation in the abundance and distribution of gases therefore arises from secondary mass-transfer and mass-transport processes, specifically molecular diffusion and exchange of gases between soil gas and pore water. Due to their inert biogeochemical behavior, mass-transport processes dominate the post-recharge evolution of NG abundances in the subsurface. Depletion and enrichment of NGs relative to air is attributed to advective transport along the total pressure gradient. However, gas advection alone cannot account for the observed elemental fractionation of NGs, which is indicative of mass-dependent variations in diffusive fluxes opposite to the total gas pressure gradient. Molecular diffusion coefficients for NGs exhibit an inverse relationship with atomic number (Jähne et al., 1987; Bourg and Sposito, 2008). The magnitude of diffusive NG fluxes, opposite to predominant advective flux, can therefore be expected to follow the general order $Ne > Ar > Kr > Xe$. Freundt et al. (2013) invoked this process to describe elemental fractionation of NGs along a total gas pressure gradient within an unsaturated soil. Results from the current study generalize the findings by Freundt et al. (2013) to systems affected by both anaerobic conditions (methanogenesis) and methane oxidation. Freundt et al. (2013) exclusively focused on aerobic respiration leading to advective O_2 ingress due to a reduction of total gas pressures as a result of O_2 consumption. In contrast, gas generation in the anaerobic methanogenic region locally causes overpressures, which lead to noble gas depletion, overlain by an aerobic oxidation zone that is characterized by underpressures, showing noble gas enrichment (Amos et al., 2005). Additional differences between these studies are attributed to the magnitude of advective fluxes along the reaction-induced total gas pressure gradient, which generated larger concentration gradients and likely supported higher diffusive fluxes in the current study. Since both biogeochemical and physical processes control advective fluxes, atmospheric NGs offer insight into mass-transfer and mass-transport and facilitate improved calibration of reactive transport models.

Reactive transport modeling was performed using the coupled MIN3P-DGM model to evaluate whether gas advection and mass-dependent fractionation during diffusion can explain the observed NG distributions and abundance. The DGM accounts for non-equimolar fluxes of individual gases due to mass-dependence of molecular diffusion coefficients. The MIN3P-DGM model was used to reproduce vertical concentration profiles for atmospheric NGs (Ne, Ar, Kr, Xe) and major component gases (CH_4, CO_2, O_2 and N_2). The model assumed atmospheric conditions at the ground surface, a no flux boundary condition at the water table, and made use of binary diffusion coefficients accounting for the mass-dependence of diffusive noble gas migration (see Supplementary data). The model computed the partial pressures of all gases along the depth profile based on a rate of CO_2 and CH_4 generation at the base of the domain, which was defined by the total carbon efflux measured at the ground surface (Sihota et al., 2011; Sihota and Mayer, 2012). This led to a highly constrained model and calibration was obtained by only adjusting the permeability and soil hydraulic function parameters of the sediments within the range observed at the site (Dillard et al., 1997), consistent with other previous simulations (Molins et al., 2010; Sihota and Mayer, 2012). Results of the one-dimensional simulations generally provided strong agreement with field observations (Fig. 7), both for atmospheric NGs (Ne, Ar, Kr, Xe) and major component gases (CH_4, CO_2, O_2 and N_2). The model successfully simulated both the depletion and elemental fractionation of NGs associated with the zone of CH_4 production. In this region, maximum depletions for Xe, Kr, Ar and Ne could be reproduced with deviations of <10% in comparison to observed data (Fig. 7). Remaining differences between observed data and simulation results, in particular in the methanotrophic region, are likely due to a combination of processes involving local permeability reduction, possibly caused by biofilm growth, and/or multidimensional effects not captured by the current model.

Modeled gas fluxes also support the conceptual model for biogenic gas production and transport developed by Amos et al. (2005). Simulations indicate that diffusive fluxes dominate O_2 and CH_4 transport to the reaction zone from the ground surface and methanogenic zone, respectively. Advective fluxes represent less than 10% (O_2) and 20% (CH_4) of the net flux due to large concentration gradients that develop between the methanotrophic zone and the ground surface or methanogenic zone, respectively. In contrast, upward transport of CO_2 from the methanogenic zone is dominated by advection along the reaction-induced total pressure gradient. Closer to the ground surface, diffusion becomes the dominant mechanism of CO_2 transport due to a decrease in the total pressure gradient above the zone of CH_4 removal. The model also demonstrated that mass-dependent variations in diffusive fluxes among NGs developed opposite to the total pressure gradient. These results suggest that the observed distribution and abundance of NGs within the vadose zone results from the net influence of advective and diffusive transport between zones

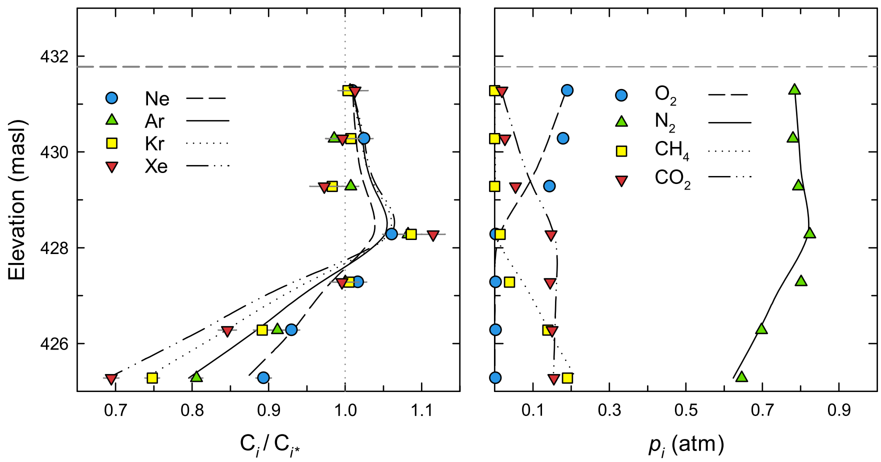

Fig. 7. Profiles of measured (symbols) and simulated (lines) gas abundances at vapor well 601G. Noble gas abundances are normalized to atmospheric values (left), whereas major component gases (right) were calculated as partial pressures.

of CH_4 production and removal by oxidation. Both field observations and modeling results clearly indicate that Kr and Xe are the most sensitive noble gas tracers for identifying biogeochemical zones and transport processes in the vadose zone (Fig. 7).

5.2. Gas dynamics in the saturated zone and mass transfer to the vadose zone

Extensive depletion of atmospheric NGs was observed for several wells located within the dissolved hydrocarbon plume at the Bemidji site. Additionally, mass-dependent fractionation of NGs was observed with depletion generally exhibiting an inverse relationship with atomic mass (i.e., $\Delta Ne > \Delta Ar > \Delta Kr > \Delta Xe$). The depletion of NGs relative to ASW is indicative of gas exsolution leading to bubble formation and possibly mass transfer of gases to the vadose zone via ebullition. The observed trends in NG abundances are consistent to those reported by Brennwald et al. (2005) for ebullition in methanogenic lake sediments. Bubble formation is associated with methanogenic degradation of organic matter due to the relatively low solubility of CH_4 in water (Martens and Val Klump, 1980; Chanton et al., 1989). Atmospheric NGs present in groundwater partition into these bubbles by diffusion across the gas–water interface. Assuming kinetic limitations are negligible, NG partitioning will proceed until solubility equilibrium is achieved. The assumption of equilibrium partitioning between pore water and entrapped gas bubbles is justified considering that equilibration is achieved on the order of hours as suggested by Holocher et al. (2002), while gas generation via contaminant degradation is a slow process that has evolved continuously over the past 20 years at this site. In contrast to atomic diffusion coefficients, NG solubilities exhibit a positive relationship with atomic mass. Equilibrium partitioning into CH_4 bubbles will therefore cause preferential depletion of lighter NGs, which is consistent with our observations.

Geochemical modeling was performed to examine NG evolution during CH_4 production and the observational data was correlated with models that simulate (i) gas exsolution and bubble entrapment and (ii) gas exsolution, bubble entrapment, and in addition periodic bubble release to the vadose zone via ebullition (see Supplementary data for model setup). Both models neglected the influx of noble gases from upgradient regions via lateral groundwater flow, recharge, or via upwards diffusion from the uncontaminated aquifer below. This simplification is justified, because quantitative data is not available on the magnitude and distribution of these fluxes due to aquifer heterogeneities, and the presence of free phase oil and entrapped gas bubbles. The negligible contribution of vertical diffusion from the aquifer below was supported by the observed $^{20}Ne/^{22}Ne$ and $^{36}Ar/^{40}Ar$ ratios, ranging from 9.74 to 9.94 and 0.0033 to 0.0034, respectively (Fig. 8), consistent with findings of Brennwald et al. (2005). The noble gas isotope ratios are generally clustered around the isotopic signature of the

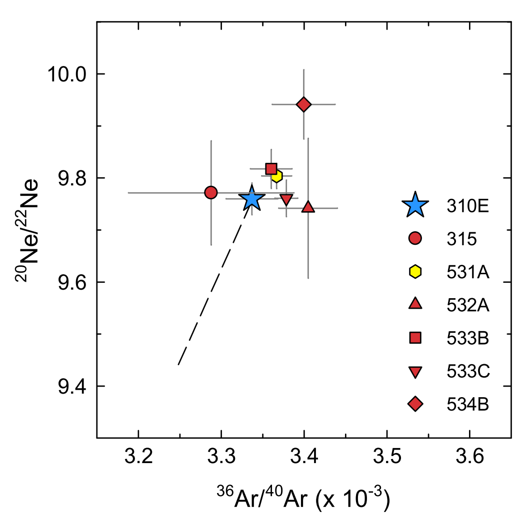

Fig. 8. Isotopic ratios of Ne and Ar in water samples from saturated-zone wells. Dotted line represents expected isotopic ratios based on the Rayleigh equation using the signature for 310E to represent no fractionation.

background well (310E), where gas exsolution and ebullition are assumed negligible (Fig. 8). Observed Ne/Xe ratios were correlated to the volume of gas generated as predicted by both models (Fig 9a and b), while noble gas ratios for Ne/Kr and Kr/Xe, as well as the noble gas concentrations for Ne, Kr and Xe (Fig 9c and d) were plotted for the same degree of reaction progress for each of the samples. Fig. 9 shows that both models can explain the data equally well. With the exception of data collected for wells 531A (which displayed an excess air signature and is not shown), the model effectively represented the evolution of NGs from the groundwater in equilibrium with the atmosphere (ASW). The fact that the five independent parameters (Ne, Kr, Xe, Ne/Kr and Kr/Xe) also follow the simulated compositional evolution suggests that both models are internally consistent. This finding implies that ebullition is possibly occurring at the Bemidji site; however, it cannot be confirmed with certainty based on the current data set. Nevertheless, for the case of gas exsolution and entrapment the results suggest that during the biogeochemical evolution up to 40 ml of gas have been generated per liter of groundwater (Fig. 9a and c), while the case including ebullition suggests historic gas production of up to 20 ml (Fig. 9b and d). This is likely a conservative estimate, as noble gases may be replenished by lateral groundwater flow or recharge from above. In addition, if noble gases are also replenished via vertical diffusion (as suggested by Bourg and Sposito, 2008), the observed noble gas signature would further underestimate the degree of gas generation.

The models were extended for higher degrees of gas generation (Fig. 10) to evaluate how the occurrence of ebullition, in the absence of advective and diffusive gas replenishment, affects the evolution of pore gas composition. This modeling exercise indicates that periodic gas removal via ebullition leads to a rapid decline in noble gas ratios, in particular the ratios of Ne/Kr and Ne/Xe decline quickly and become less than unity for moderate volumes of gas generation (<100 ml L^{-1} H_2O for Ne/Kr, Fig. 10). This behavior is not observed when gas bubbles remain entrapped. In addition, only for the model with ebullition, the Ne/Xe ratios become lower than the Kr/Xe ratios, also for volumes of gas generation <100 ml L^{-1} H_2O. Although partitioning of NGs into entrapped gas bubbles in the absence of ebullition also results in a decrease of dissolved concentrations, the decreases are substantially smaller and a reversal of noble gas ratios does not occur. These

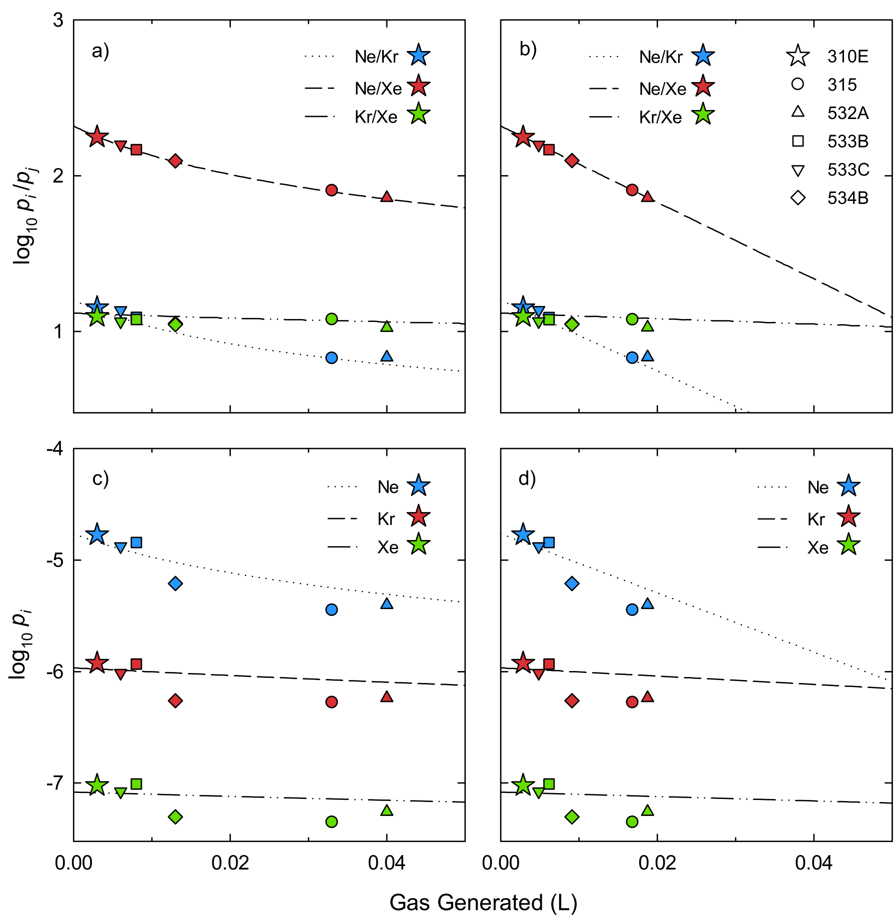

Fig. 9. Observed and simulated noble gas ratios (panels a and b) and partial pressures (panels c and d) of dissolved noble gases plotted versus cumulative volume of gas produced. Observed data was correlated with models that simulate (i) gas exsolution and bubble entrapment (panels a and c) and (ii) gas exsolution and ebullition (panels b and d). Measured atmospheric noble gas abundances were used as the initial condition for these simulations.

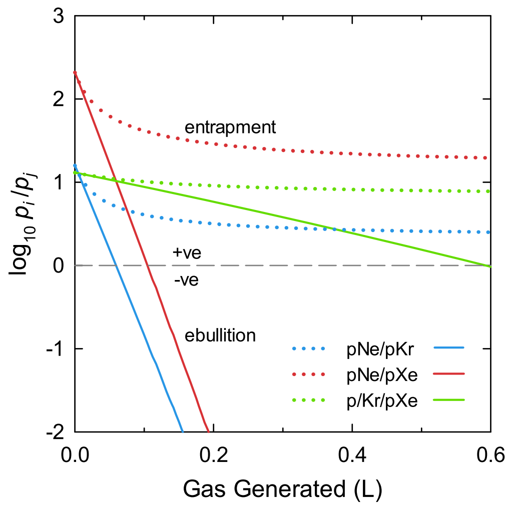

Fig. 10. Direct comparison of models that simulate (i) gas exsolution and bubble entrapment (dotted lines) and (ii) gas exsolution and ebullition (solid lines) for an extended range of gas formation. Measured atmospheric noble gas abundances were used as the initial condition for these simulations.

observations suggest that it may be possible to use noble gas concentrations and ratios to prove the occurrence of ebullition in the field at sites with a higher degree of biogenic gas generation. Furthermore, the study results suggest that Ne/Kr and Ne/Xe ratios are particularly sensitive indicators of the ebullition process.

6. CONCLUSIONS

Atmospheric noble gases (NGs) have provided unique insight into biogenic gas dynamics associated with methanogenic degradation of hydrocarbons in a shallow unconfined aquifer. Noble gas signatures were used to trace biogeochemical and transport processes both in the vadose zone and the saturated zone. In the vadose zone, the deviation of NGs relative to atmospheric values was associated with advective gas transport along a total pressure gradient. This gradient was established between spatially separated zones of methanogenesis and methane oxidation associated with hydrocarbon degradation. Reactive transport simulations confirmed that observed elemental fractionation of NGs resulted from mass-dependent diffusive fluxes opposite to the total pressure gradient. These results demonstrate that heavy noble gases (i.e., Kr and Xe), which exhibit lower diffusion coefficients, are the most sensitive tracers to delineate reaction zones and identify advective gas transport in the vadose zone. In the saturated zone, depletion of NGs relative to air-saturated water (ASW) was observed for wells located within the dissolved hydrocarbon plume and near the water table. Mass-dependent elemental fractionation of NGs was attributed to equilibrium partitioning of NGs into methane bubbles. Modeling provided conservative estimates of historical gas generation via biodegradation and further demonstrated that ratios of Ne/Kr and Ne/Xe may serve as effective indicators of ebullition at sites with high rates of biogenic gas generation.

These results provide important constraints for carbon cycling in shallow unconfined aquifers, including the release of CO_2 and CH_4 from the groundwater zone to the overlying vadose zone. These results are not only applicable to contaminated groundwater systems, but should also be relevant to other partially water saturated systems that are affected by biogeochemical reactions, including bogs, peats, and wetlands.

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APPENDIX A. SUPPLEMENTARY DATA

Supplementary data associated with this article can be found, in the online version, at http://dx.doi.org/10.1016/j.gca.2013.12.008.

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